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INTERFERENCE CONTROL IN CHILDREN WITH AND WITHOUT ADHD: A SYSTEMATIC REVIEW OF FLANKER AND SIMON TASK PERFORMANCE

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The present review systematically summarizes the existing research that has examined two reaction-time-based interference control paradigms, known as the Eriksen Flanker task and the Simon task, in children with and without ADHD. Twelve studies are included, yielding a combined sample size of 272 children with ADHD (M age 9.28 yrs) and 280 typically developing children (M age 9.38 yrs). As predicted, specific disadvantages were found in the ADHD group in terms of reaction time, percentage of errors, and efficiency of performance on incongruent relative to congruent trials, providing evidence for weaker interference control in this group.

Keywords: ***Review; ADHD; Attention; Interference control; Children.***

A very substantial amount of research has been directed at identifying the neurocognitive processes responsible for the inattentive, hyperactive, and impulsive behaviors observed in children with attention deficit/hyperactivity disorder (ADHD). A number of influential theories have emerged, each of which has emphasized a different cognitive process or combination of processes as the most likely candidate(s). Of these many potential cognitive processes, a large body of evidence points to impairments in executive functions, especially aspects of inhibition (e.g., Oosterlaan, Logan, & Sergeant, 1998; Pennington & Ozonoff, 1996; Willcutt, Doyle, & Nigg, 2005).

Various taxonomies of inhibition have been proposed and have been applied to both children and adults (e.g., Bjorklund & Harnishfeger 1990; Friedman & Miyake, 2004; Nigg, 2000). While terminology varies across classification system, a distinction is consistently made between the ability to prevent or to suppress a highly practiced automatic or dominant response and the ability to filter out competing information that is irrelevant to

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the task being performed. The former has been referred to by terms such as prepotent response inhibition (Barkley, 1997; Friedman & Miyake, 2004), behavioral inhibition (Nigg, 2000), and response suppression (Nigg, 2006); while the latter has been labelled as resistance to distractor interference (Friedman & Miyake, 2004), conflict resolution (Posner & DiGirolamo, 1998), executive attention (Posner & Rothbart, 2007), and interference control (Nigg, 2000). We will refer to these two processes by the terms *response suppression* and *interference control,* respectively. While separate constructs, these processes are related to some extent (Friedman & Miyake, 2004; Harnishfeger, 1995) and both are needed for the deliberate, effortful control of behavior towards future goals (Barkley, 1999). The current paper will focus on interference control in children with and without ADHD.

In the ADHD literature, response suppression has most frequently been measured with the Stop task (Logan, 1994) while interference control has been primarily assessed with the Stroop Color Word task (Stroop, 1935). Deficits in response suppression, as measured by the Stop task, have been widely reported in children with ADHD (Pennington & Ozonoff, 1996; Sergeant, Geurts, & Oosterlaan, 2002; Willcutt, Doyle, Nigg, Faraone, & Pennington, 2005). An impairment in this domain has been described as one of the most reliably replicated findings in the literature on the neuropsychological functioning of children with ADHD (Willcutt et al., 2005).

Recently, the construct of interference control has received more attention in the literature as theorists have sought to quantify the behavioral symptoms of ADHD within a well-validated cognitive neuroscience model of attention (e.g., Berger & Posner, 2000; Swanson et al., 1998). This model, known as the Attention Network Theory (Posner & Petersen, 1990), describes three neural networks of attention. These include the alerting network (i.e., the achievement and maintenance of an optimally alert attentional state), the orienting network (i.e., the movement of visual attention in space), and the executive network. The executive network is associated with the control of goal-directed behavior, inhibition, and conflict resolution (Berger & Posner). Interference control is one of the key functions of the executive network (Posner & DiGirolamo, 1998). According to proponents of the Attention Network Theory, behavioral and imaging studies have provided support for the hypothesis that aspects of executive control are impaired in ADHD (Berger & Posner), but it has been argued that results are inconclusive with regard to interference control more specifically (Nigg, 2006).

The majority of research that has examined interference control in an ADHD population has employed the Stroop task. Several variants of this task exist and most consist of three parts. First, participants complete the *word task* in which they are asked to name a series of color words (e.g., read the word "red," "blue," etc., written in black ink). Next is the *color task* in which they are to name the color of a bar of X's (e.g., XXXXX in red ink). Finally, in the *color-word task* participants are shown the names of colors printed in conflicting ink colors (e.g., the word "blue" printed in red ink). The color-word task is thought to measure interference control ability because it requires participants to inhibit the dominant response tendency to read the words and, instead, name the color of the words (Homack & Riccio, 2004). Several different scoring methods have been used to derive an interference score but all involve some comparison between performance on the control task(s) (color task, word task) relative to performance on the color-word task (Homack & Riccio; Lansbergen, Kenemans, Verbaten, & van Engeland, 2007).

To date, four research teams have conducted meta-analytic reviews of the performance of children and/or adults with and without ADHD on the Stroop task. Two meta-analyses found moderate effect sizes (*d*) of .69 (Pennington & Ozonoff, 1996) and .75 (Homack & Riccio, 2004), in which children with ADHD demonstrated weaker interference control abilities. One reported a nonsignificant effect size of .35 (van Mourik, Oosterlaan, & Sergeant, 2005). To clarify these inconsistent results, Lansbergen et al. (2007) conducted a fourth meta-analysis. They employed a more detailed analysis to take into account the various administration formats (i.e., set of cards vs. computerized administration) and scoring procedures that have been used. Their results indicated that effect sizes were larger for studies that used time-based dependent variables (DVs) such as time taken to read each card or reaction time (RT) per item (d = 1.11 and .55 for analyses with and without four outliers, respectively) than when studies used a frequency count as the DV (e.g., number of items completed in 45 seconds). Based on their analysis, Stroop task performance was concluded to be most reliably impaired in children and adults with ADHD when administered in a computerized format and when DVs were measured as RT per item or time taken to read each card.

It has been argued that limitations of the Stroop task itself do not allow clear conclusions to be drawn about interference control ability in children with ADHD (Nigg, 2001; van Mourik et al., 2005). Several possible explanations for inconsistent Stroop task findings in this population have been highlighted in the literature. For example, van Mourik et al. (2005) noted that Stroop task performance also depends upon the "nonexecutive" skills of rapid naming, reading speed, and reading ability and all of these have been found to be poorer in children with ADHD. Studies have not always adequately controlled for baseline performance on the word task (to control for reading ability) or the color task (to control for rapid naming ability). They also point out that a significant proportion of children with ADHD also have comorbid disorders, most notably Reading Disorders (RD). For example, children with RD may show more interference on the Stroop task if they are struggling to read the color words. Stroop task studies have not always sufficiently controlled for the presence or severity of comorbid disorders. A third possible explanation is the fact that there are a variety of different methods for quantifying interference on the Stroop task (Lansbergen et al., 2007). As noted above, Lansbergen et al. (2007) found different results and effect sizes depending upon the scoring method and DVs used. Fourth, Stroop interference effects have been found to vary in magnitude depending on how frequently incongruent trials are presented (Botvinick, Braver, Yeung, Ullsperger, Carter, & Cohen, 2004) and that may have presented a confounding variable across studies.

Despite the limitations described here, studies of the Stroop task have made a significant contribution to our current understanding of interference control in ADHD. There is some evidence to suggest that children with ADHD may have weaker interference control from studies that have used alternative computerized measures of this ability (e.g., Cornoldi et al., 2001; Crone, Jennings, & van der Molen, 2003; Konrad, Neufang, Hanisch, Fink, & Herpertz-Dahlmann, 2006) and/or functional magnetic resonance imaging (Bush et al., 1999; Konrad et al., 2006). Among the various paradigms that have been used, two experimental tasks drawn from the cognitive literature may be particularly useful for quantifying interference control in children with ADHD. The Eriksen Flanker task (Eriksen & Eriksen, 1974) and the Simon task (Simon, 1990) are nonverbal computerized tasks that precisely measure interference control using RT.

The Flanker task is widely used to measure interference control (e.g., Fan, Flombaum, McCandliss, Thomas, & Posner, 2003). This task calls for the participant to identify a target item defined by its location (often, but not exclusively presented at fixation) while ignoring one or more distracting items that flank the target and whose identities may activate the correct (on congruent trials) or incorrect (on incongruent trials) response. This effect is illustrated here using the Flanker task from the Attention Network Test (ANT; Fan, McCandliss, Sommer, Raz, & Posner, 2002). In this task, a set of five arrows is presented on a computer screen and participants are asked to indicate the direction of the central arrow (target) that is pointing to the left or to the right with a speeded keypress response. The target is flanked by two identical arrows on either side (distractors) that are either pointing in the same direction (known as a *congruent* trial, →→→→→) or the opposite direction of the target arrow (known as an *incongruent* trial, →→←→→). On incongruent trials, the irrelevant, distracting information from the flanking arrows must be filtered out, a process thought to require executive control (e.g., Posner & DiGirolamo, 1998). The DVs derived from this task are RT and accuracy. The general finding of longer RT on incongruent trials indicates that additional attentional processing is required to filter out the distracting information. Similarly, accuracy tends to be lower on incongruent trials providing additional evidence that the cognitive processing required for these trials is more effortful (for a review of this these effects in children see Ridderinkhof & van der Stelt, 2000).

In a Simon task, a single item is presented in one of two locations (to the left or right of fixation). The participant must indicate the identity of the item with a left or right keypress. Because of the natural tendency to respond in the direction of task-relevant stimuli, RT is faster when the required response (e.g., right keypress) is congruent with the location of the stimulus (e.g., right side of fixation). For example, in the Simon task taken from Zimmerman and Fimm’s (2002) Test of Attentional Performance, participants are asked to make a left keypress when they are presented with an arrow pointing to the left and a right keypress when they are presented with an arrow pointing to the right. The arrows are presented on the left and right side of fixation on the computer screen. Response latencies to a right pointing arrow are faster when that stimulus is presented on the right and slower when it is presented on the left (Lu & Proctor, 1995).[1]

On both Flanker and Simon tasks, the amount of interference displayed by an individual is quantified in terms of a difference score (often referred to as the “congruency effect”). To calculate the congruency effect, the mean RT (or % correct) on congruent trials is subtracted from the mean RT (or % correct) on incongruent trials. Larger scores are indicative of less efficient interference control. These difference scores control for the large individual differences in speed of responding when RT is the DV. Without such a subtraction, a high or low score could be attributed to the participant simply being a slow or fast responder. When making comparisons at the group level, potential differences in interference control ability can be evaluated by examining the main effect of group on the RT or accuracy congruency effects. Group performance may also be reported in terms of mean RT or accuracy on incongruent and congruent trials separately. In this situation, the group-by-trial type (i.e., incongruent vs. congruent) interaction effect provides the necessary information to draw conclusions about group differences in the size of the congruency effect.

In a choice reaction time task, differences in the quality of performance as manifested by speed of responding and/or accuracy of responding may occur as a function of factors such as trial type and group as well as interactions between these factors. For example, one group may be faster but more inaccurate responders while another may be slower but more accurate. In this case, considering RT or accuracy in isolation does not

[1]This version of the Simon task is sometimes referred to as a Spatial Stroop Task.

fully reflect performance efficiency. Thus, when scholars are interested in the overall efficiency of performance, it is desirable to use a measure that combines speed and accuracy (R. M. Klein, Christie, & Ivanoff, 2004). One such measure, RT divided by proportion correct, was first put forward by Townsend and Ashby (1983) and subsequently called Inverse Efficiency (IE) by Christie (1995). It was given this name because less efficient overall performance is reflected in larger IE scores. Inverse Efficiency has also been used to measure overall quality of performance in a study of covert orienting and attention filtering in typically developing children (Akhtar & Enns, 1989).

Flanker and Simon tasks offer a number of advantages for the study of interference control in children with ADHD. Firstly, these computerized tasks are theoretically driven and quantify interference effects precisely in terms of RT and accuracy. This allows for careful control and manipulation of task and measurement variables. Secondly, like the Stroop task, the Flanker and Simon tasks are valid measures of activity in the anterior cingulate cortex, the brain area thought to be responsible for interference control and conflict monitoring (Fan et al., 2003). Thirdly, they do not require that the participants be able to read or to make any verbal response. This eliminates the potential confounds of reading-related abilities noted on the Stroop task. Most importantly, these tasks may provide converging evidence to support the conclusion that interference control is impaired in ADHD.

The present review summarizes the available literature that has used a Flanker and/or Simon task to assess interference control in children with and without ADHD. More specifically, to determine whether children with ADHD show larger congruency effects than their typically developing (TD) peers. We were interested in the main effect of group when a study reported results in terms of difference scores or the interaction between group (i.e., ADHD vs. TD) and trial type (i.e., incongruent vs. congruent) when a study reported results for incongruent and congruent trials separately. Three DVs were evaluated: RT, accuracy (% error), and IE. In line with the predictions of Berger and Posner (2000) and the reviews by Pennington and Ozonoff (1996), Homack and Riccio (2004), and Lansbergen et al. (2007), we hypothesized that children with ADHD would show more interference (as in larger RT, accuracy or IE congruency effects) than their TD peers. A traditional (e.g., Rosenthal, 1991) quantitative meta-analytic approach could not be used because some data were presented in such a manner that it was not possible to find the appropriate *F*-values or to calculate the measures of effect size needed to conduct such an analysis. Consequently, qualitative, graphical, and nonparametric statistical methods are used to summarize the data across studies.

METHOD

Literature Search

We conducted a literature search to locate studies published in peer-reviewed journals and unpublished dissertations in which a Flanker and/or a Simon task were employed with a sample of children with ADHD and a group of TD children. The following strategies were used to locate studies for this review. PsychInfo and Medline Databases were searched for all articles published between January, 1980 and December, 2007. The earliest date was selected because it was the year that the *Diagnostic and Statistical Manual of Mental Disorders*, third edition (*DSM-III*; American Psychiatric Association, 1980) was introduced and earlier conceptualizations of the disorder were deemed sufficiently dissimilar as to prevent meaningful comparisons. The following search terms were used in various combinations: ADHD, child, interference control, attention network, attentional performance, executive attention, executive control, Flanker, Simon, conflict, incompatibility, focused attention, selective attention, inhibition, and visual attention. The search was limited only by age (6 to 17 years) and language (English). We further reviewed the reference lists of the articles that met these criteria to reduce the possibility that a relevant study was overlooked.

Study Selection

A study was included if: (1) there was a group of children who met diagnostic criteria for ADHD according to the *DSM-III* (American Psychiatric Association, 1980), *DSM-III-R* (American Psychiatric Association, 1987), or *DSM-IV* (American Psychiatric Association, 1994) and a comparison group of TD children; and (2) the study employed a computerized, nonverbal Flanker and/or Simon task in which the children were required to respond with speeded keypresses. We located nine studies that employed a Flanker-type task, seven of which met our inclusion criteria (Booth, Carlson, & Tucker, 2007; Crone et al., 2003; Jonkman et al., 1999; Konrad et al., 2006; McLaughlin, 2002[2]; Scheres et al., 2004; Vaidya et al., 2005; see Table 1). One study was not included because the task format was not computerized and involved sorting decks of cards (Hooks, Milich, & Lorch, 1994). A second was not included because an analysis of task validity found that the experimenter-developed task did not produce the expected congruency effect (i.e., the main effect of trial type on RT was not significant) in the control group (Sparkes, 2006). We located seven studies that employed a Simon task and five met inclusion criteria (Dreschler, Brandeis, Foldenyi, Imhof, & Steinhausen, 2005; McLaughlin, 2002; Sparkes, 2006[3]; Tsal, Shalev, & Mevorach, 2005; Tucha, Prell, et al., 2006; see Table 1). Of the two studies that were not included, one did not have a control group (Hanisch, Konrad, Gunther, & Herpertz-Dahlmann, 2004) and one required that the children read a word presented on a computer screen in order to know which button to push (Zang et al., 2005). When more than one article reported on the same data set, only one of the studies was included in the review. The data reported in Castellanos, Sonuga-Barke, Scheres, Di Martino, Hyde, and Walters (2005), Ridderinkhof, Scheres, Oosterlaan, and Sergeant (2005), and in Scheres et al. (2003) were represented in Scheres et al. (2004). The data from Tucha, Walitza, et al. (2006) were represented in Tucha, Prell, et al. (2006). The data from Konrad, Neufang, Fink, and Herpertz-Dahlmann (2007) were represented in Konrad et al. (2006) and the data from Jonkman, van Melis, Kemner, and Markus (2007) were represented in Jonkman et al. (1999).

One study reported only the overall RT and error rates (Tucha, Prell, et al., 2006) and that prevented comparisons between the two trial types. Therefore, this study was excluded from the statistical analyses due to the limited amount of information that could be obtained.[4] The 12 studies included in the review yielded a combined sample size of 272 children with ADHD (*M* age 9.28 yrs) and 280 typically developing children (*M* age 9.38 yrs). Sample-related variables (e.g., participant selection and diagnostic procedures, ADHD

[2]McLaughlin's (2002) study measured interference control using a combined Flanker and Simon task but the results were reported separately for each task.

[3]Sparkes' (2006) dissertation included a Flanker and a Simon task but only the results for the Simon task will be reported here.

[4]Both corresponding authors for the Tucha, Prell, et al. (2006) study were contacted three times each via email but we did not receive a reply.

Table 1 Participant Selection Criteria

Study	Group	*N* & Sex	Age	IQ	Comorb.[a]	Hrs off Med.[b]	Pop.[c]	Exclusions	*DSM* Version/ ADHD Diagnostic Procedures & Measures
Flanker									
Booth et al. (2007)	ADHD[d] C = 16 HI = 0 I = 26	31M 11F	9.6 (1.8)	110 (15)	14/42 ODD 3/42 LD	18	1	IQ <80, neurological disorder, psychosis, psychoactive medications other than stimulants	*DSM-IV* Previous clinical diagnosis DSM-IV Checklist (P,T)
	TD	12M 12F	9.5 (1.6)	120 (11)	1/24 ODD				
Crone et al. (2003)	ADHD TD	17M 5F 13M 9F	9.0 (1.9) 9.0 (1.8)	91 (14.2) 99 (16.8)	NR	24	1	IQ <70, neurological disorder, medical disorder, CD, LD	*DSM-III-R* Previous clinical diagnosis using structured parent interview and teacher ratings, IOWA-C (P), CBCL
Jonkman et al. (1999)	ADHD TD	13M 1F 12M 2F	9.5 (2.2) 10.5 (1.4)	98 (12.0) 109 (12.6)	NR	72	1	IQ <80	*DSM-III-R* Previous clinical diagnosis, CTRS, CBCL
Konrad et al. (2006)	ADHD C = 9 HI = 1 I = 6	16M 0F	10.2 (1.9)	103 (12)	5/16 ODD 3/16 AD	Stimulant naïve	1	IQ <80, Previous treatment with stimulant or psychoactive medications, psychosis, receptive language disorder, mania, MDD, substance abuse, PDD	*DSM-IV* K-SADS, German Parent and Teacher Report of ADHD Symptoms
	TD	16M 0F	10.1 (1.3)	105 (10)	2/16 ODD 3/16 AD				

(Continued)

Table 1 *(Continued)*

Study	Group	*N* & Sex	Age	IQ	Comorb.[a]	Hrs off Med.[b]	Pop.[c]	Exclusions	*DSM* Version/ ADHD Diagnostic Procedures & Measures
McLaughlin (2002)	ADHD-C TD	13M 3F 18M 6F	11.2 (1.9) 11.4 (1.3)	104.8 (13.1) 115.3 (18.3)	3/16 LD	24	2	IQ <80, ADHD-I, neurological disorder, TS	*DSM-IV* Previous diagnosis of ADHD, DBD, IRS, IOWA-C (P,T), CBCL, TRF
Scheres et al. (2004)	ADHD C = 14 HI = 1 I = 8	23M 0F	8.7 (1.7)	97.6 (14.7)	11/23 ODD 1/23 CD	Stimulant naïve	1	IQ <70, Previous treatment with stimulant medication	*DSM-IV* DBD, DSM-IV Screen
	TD	22M 0F	9.6 (1.8)	104.7 (19.1)					
Vaidya et al. (2005)	ADHD-C TD	7M 3F 7M 3F	8.8 (0.9) 9.2 (1.3)	104.3 (16) 128.4 (16.7)	NR	36	2	IQ <85, neurological disorder, affective disorder, language disorder	*DSM-IV* DISC, SNAP (P,T), CBCL, TRF
Simon									
Dreschler et al. (2005)	ADHD	26M 2F	11.0 (1.7)	99.0 (10.5)	NR	48	2	IQ <80	*DSM-III-R*
(Time 1 data only)	TD	19M 6F	10.6 (1.2)	105 (9.8)					
McLaughlin (2002)	ADHD-C TD	13M 3F 18M 6F	11.2 (1.9) 11.4 (1.3)	104.8 (13.1) 115.3 (18.3)	3/16 LD	24	2	IQ <80, ADHD-I, neurological disorder, TS	*DSM-IV* Previous diagnosis of ADHD, DBD, IRS, IOWA-C (P,T), CBCL, TRF

Sparkes (2006)	ADHD C = 14 HI = 1 I = 4	13M 6F	9.8 (1.3)	100 (11.9)	9/19 ODD	12	2	IQ <80, neurological disorder, CD	*DSM-IV* Previous diagnosis of ADHD, KSADS, CPRS, DBD
	TD	14M 12F	9.7 (1.4)	115 (13.6)					
Tsal et al. (2005)	ADHD C = 20 HI = 0 I = 7	20M 7F	8.2 (1.2)	NR	8/27 LD	Off for testing	NR	IQ <80, neurological disorder, mood disorder, AD, CD	*DSM-IV* Previous clinical diagnosis, ADHD-IV Rating Scale (P)
	TD	6M 9F	8.0 (0.6)						
Tucha, Prell, et al. (2006)	ADHD C = 28 HI = 25 I = 5	49M 9F	10.8 (0.3)[e]	98.1 (1.5)[e]	Children with other Axis I disorders were excluded	Placebo condition	1	IQ <85, Other Axis I disorder	*DSM-IV* Previous clinical diagnosis
	TD	49M 9F	10.8 (0.3)[e]	Not assessed					

Note: [a]Comorbidity; [b]Number of hours off stimulant medication for testing, [c]Population from which children with ADHD were recruited (1 = psychiatric/neurological clinics; 2 = general population; 3 = summer research and treatment program for children with Disruptive Behavior Disorders); [d]Data for ADHD-Combined Type and ADHD-Predominately Inattentive Type was combined into one ADHD group for the purpose of this review; [e]Standard Error of Measurement. Abbreviations: AD = Anxiety Disorder; ADHD = Attention Deficit/Hyperactivity Disorder; C = ADHD-Combined Type; HI = ADHD-Predominately Hyperactive-Impulsive Type, I = ADHD-Predominately Inattentive Type; CBCL = Child Behavior Checklist; CD = Conduct Disorder; CPRS = Conners Parent Rating Scale; CTRS = Conners Teacher Rating Scale; DBD = Disruptive Behavior Disorders Scale; DBI = Disruptive Behavior and Inattention Scale; DISC = Diagnostic Interview Schedule for Children; *DSM = Diagnostic and Statistical Manual*; DSMD = Devereaux Scales of Mental Disorders; IOWA-C = IOWA Conners Scale; IRS = Impairment Rating Scale; LD = Learning Disability; MDD = Major Depressive Disorder; NR = information not reported in study; ODD = Oppositional Defiant Disorder; P = parent form; PDD = Pervasive Developmental Disorder; T = teacher form; TS = Tourette's Syndrome; SNAP = Swanson, Nolan, and Pelham Teacher and Parent Rating Scale; TRF = Teacher Report Form.

Table 2 Task Variables

Study	Task	Target[a]	Distractors[b]	Total Practice Trials	Number of Flanker or Simon Trials	Total Experimental Trials	Total Time on Task[c]
Flanker							
Booth et al. (2007)	ANT (Child)	Fish pointing left or right	Fish pointing left or right	24	144	144	15
Crone et al. (2003)	Go/No-Go Flanker Task	Arrow pointing left or right	Arrows pointing left or right	40	480	1080	36.5
Jonkman et al. (1999)	Flanker Task	Arrow pointing left or right	Arrows pointing left or right	32	480	480	30*
Konrad et al. (2006)	Modified ANT (Adult)	Arrow pointing left or right	Arrows pointing left or right	"5 min training session"	NR	NR	NR
McLaughlin (2002)	Combined Flanker/Simon	Number 1, 2, 3, or 4	Number 1, 2, 3, or 4	30	30	240	25
Scheres et al. (2004)	Flanker Task	Arrow pointing left or right	Arrows pointing left or right	180	360	360	27*
Vaidya et al. (2005)	Go/Nogo Flanker Task	Arrow pointing left or right	Arrows pointing left or right	"10 to 20 practice trials"	156[d]	208[d]	12*
Simon							
Dreschler et al. (2005)	TAP Incompatibility Task	Arrow pointing left or right[e]		NR	60	140	NR[f]
McLaughlin (2002)	Combined Flanker/Simon	Number 1, 2, 3, or 4[e]		30	30	240	25
Sparkes (2006)	Simon Task	Heart or diamond[g]		36	216	216	NR[f]
Tsal et al. (2005)	Stroop-like Task Direction Condition	Arrow pointing up or down		10	NR	NR	NR
Tucha, Prell, et al. (2006)	TAP Incompatibility Task	Arrow pointing left or right[e]		"a brief sequence of practice trials preceded each test"	57	1187	NR[f]

Note. [a]Targets were presented above or below fixation unless otherwise noted; [b]Distractors were presented on the left and right side of the target, unless otherwise noted; [c]Times with an asterisk were estimated by the authors based on the number of experimental and practice trials and the reported duration of each trial; [d]Number of trials was reported as a range so the mean number of trials was used in these calculations; [e]Target presented on left or right of fixation; [f]Duration of each trial not reported therefore it was not possible to estimate time on task; [g]participants were instructed to press a left-sided key when a heart appeared and a right-sided key when a diamond appeared. Abbreviations: ANT = Attention Network Test; NR = information not reported in the article; TAP = Test of Attentional Performance.

subtype included, comorbidity, etc.) are summarized in Table 1 and task-related variables are summarized in Table 2.

RESULTS

Flanker and Simon studies were combined for all analyses but are denoted in Figure 1 with separate symbols (i.e., white diamonds for Flanker studies and black diamonds for Simon studies). In studies that reported congruency effects in terms of difference scores with RT and accuracy as the DV, the main effect of group was examined. In studies that reported these variables separately by trial type, we examined the group-by-trial type interaction effect. Low sample sizes, variability in methodology, and missing information precluded statistical combination using classical meta-analytic techniques. We therefore employed an alternative graphical method for representing this data following the general approach developed by Mullane and Klein (2008). The group differences in interference control ability are summarized across studies in Figure 1 by RT (Figure 1a), % correct (Figure 1b), and IE (Figure 1c).

To create data points that captured the group differences in interference control, we first obtained the difference scores (RT and % correct) for the ADHD and TD groups. This information was obtained either directly from the tables/text or was calculated with the following formula: ([*M* score on incongruent trials] – [*M* score on congruent trials]). Larger numbers are indicative of greater interference and therefore poorer interference control. Secondly, an IE difference score was calculated for each group in each study that reported both DVs (RT and accuracy) using the formula: ([*M* RT on incongruent trials/ *M* error on incongruent trials] – [*M* RT on congruent trials/ *M* error on congruent trials]).

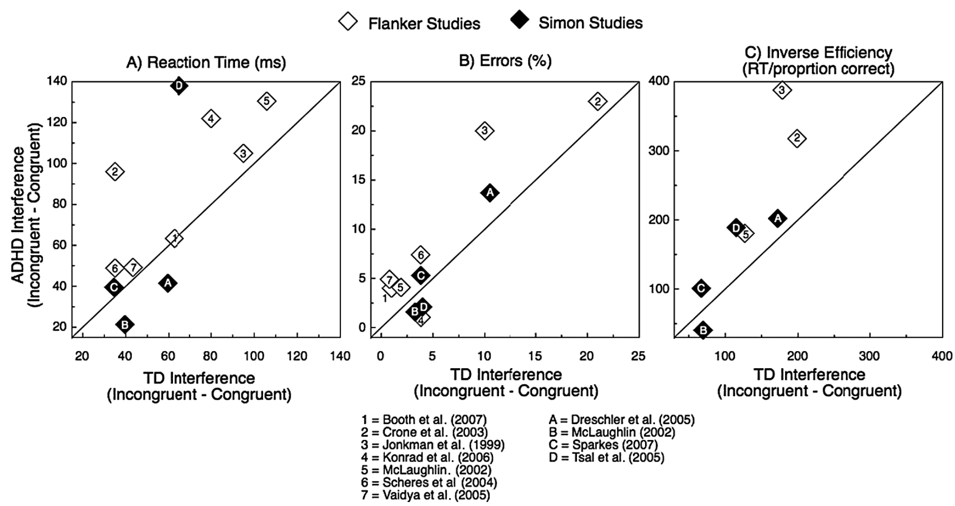

Figure 1 Relative magnitude of difference in interference between the ADHD and TD groups, as indexed by RT (a), errors (b), and IE (c). The diagonal lines represent "equal" performance for the two groups. Data points falling above the diagonal indicate weaker interference control in the ADHD group in that experiment. The further away data points fall from the diagonal line, the greater the magnitude of the difference in interference control ability between the ADHD and the TD group.

In the scatterplot depicted in Figure 1a, each data point represents the RT congruency effects for the TD children (plotted on the abscissa) and for children with ADHD (plotted on the ordinate). In this plot the diagonal line represents "equal" performance for the two groups. When a data point falls above the diagonal, the ADHD group had larger congruency effects (and therefore less efficient interference control ability) in that experiment. The further away the data point falls from the diagonal line, the greater the magnitude of the difference in interference control ability between the ADHD and the TD group. An identical procedure is used in Figure 1b for percentage of errors and Figure 1c for IE.

Finally, to statistically examine the graphical trends across studies, we conducted Wilcoxon Signed Rank tests for each DV. This test was used because the scores did not meet the criteria for parametric analyses (i.e., interval or ratio measurement, normal distribution). In this procedure, each study was considered to have two paired observations (i.e., the mean difference score for the ADHD group and the mean difference score for the TD group). Because we made a priori predictions that the ADHD group would have weaker interference control ability, as measured by each of the three DVs, p values in these analyses were one-tailed.

Reaction Time

We predicted that children with ADHD would display larger RT congruency effects than TD children. Three studies supported our predictions and found that children with ADHD showed significantly larger congruency effects (i.e., Crone et al., 2003; Konrad et al., 2006; Tsal et al., 2005). Five studies showed numerical trends in the predicted direction that did not reach statistical significance (i.e., Jonkman et al., 1999; McLaughlin, 2002, Flanker data; Scheres et al., 2004; Sparkes, 2006; Vaidya et al., 2005). One study found that the two groups were equivalent (Booth et al., 2007), two showed nonsignificant trends in the opposite direction (Dreschler et al., 2005; McLaughlin, 2002, Simon data), and one study did not provide RT separately for congruent and incongruent trials (Tucha, Prell, et al., 2006). Figure 1a illustrates these results. As predicted, the majority of data points fall above the diagonal, indicating that children with ADHD have poorer interference control relative to the TD children as indexed by RT. The Wilcoxon Signed Rank test indicated a significant difference in RT between the ADHD and TD groups across studies ($Z = -1.87$, $p = .03$, one tailed).

Accuracy

We also predicted that children with ADHD would display larger accuracy congruency effects than TD children. Two studies supported this hypothesis (Jonkman et al., 1999; Vaidya et al., 2005). Six showed a nonsignificant trend in the expected direction (Booth et al., 2007; Crone et al., 2003; Dreschler et al., 2005; McLaughlin, 2002; Flanker data; Scheres et al., 2004; Sparkes, 2006) and one did not report a statistical analysis for this specific set of values (Konrad et al., 2006). Two reported a nonsignificant trend in the opposite direction (McLaughlin, 2002, Simon data; Tsal et al., 2005). One study did not provide error rate by trial type (Tucha, Prell, et al., 2006). Figure 1b demonstrates that the majority of the data points fall above the diagonal. As predicted, the accuracy congruency effects were significantly larger for children with ADHD. Collapsed across studies, the Wilcoxon Signed Rank test indicated that this difference was statistically significant ($Z = -1.96$, $p = .03$, one tailed).

Inverse Efficiency

Inverse Efficiency scores were used to examine group differences in overall quality of performance. Because some studies did not provide the RT or the accuracies for each required combination, the number of studies for which we could compute and plot IE was limited to seven. As predicted, children with ADHD had significantly larger IE congruency effects relative to TD children. In this case, all but one of the data points in Figure 1c fall above the diagonal suggesting a clear trend for the ADHD groups to less efficient overall responders on incongruent trials relative congruent trials. Consistent with the graphical trend, the Wilcoxon Signed Rank test indicated that this difference was statistically significant ($Z = -2.20$, $p = .01$, one tailed).

DISCUSSION

The resolution of interference created when one is required to respond to one attribute of a target while ignoring distracting information that may activate a competing response is a central function of executive control (Posner & DiGirolamo, 1998). This process has recently received attention in the ADHD literature as theorists have sought to understand executive control within cognitive neuroscience models of attention such as the Attention Network Theory. To date, the vast majority of research has examined interference control using the Stroop task. Other well-established cognitive tasks of interference control such as the Flanker and Simon have been much less frequently studied. Research that has used these alternative paradigms has been marked by substantial heterogeneity in sample characteristics, methodology, and results. Consequently, this review systematically evaluated the extant literature that has examined interference control in children with ADHD using Flanker or Simon tasks.

Consistent with the general findings of the broader literature on RT-based task performance in this population (e.g., Douglas, 1999; Swanson et al., 2004), children with ADHD were generally slower to respond, made more errors and were less efficient in their performance across most task conditions. Of most importance to the present review was the examination of potential group differences in the speed, accuracy, and efficiency of processing during incongruent trials relative to congruent ones (i.e., "congruency effects"). When groups significantly differ in the size of these congruency effects, the group with the larger congruency effect is inferred to have weaker interference control. As predicted, children with ADHD were significantly more vulnerable to the interference and associated additional processing demands during incongruent trials, when the findings from all the studies satisfying our inclusion criteria were examined together. This provides further evidence that children with ADHD do in fact have weaker interference control. Our results are consistent with the majority of the reviews of Stroop task performance that have also concluded that interference control is impaired in ADHD (Homack & Riccio, 2004; Lansbergen et al., 2007; Pennington & Ozonoff, 1996). The present review extends the literature by demonstrating that an interference control deficit is also found when alternate measures of this ability are employed, notably measures that are unlikely to be so directly influenced by verbal fluency as the Stroop effect.

An overall pattern of more errors on RT-based tasks in children with ADHD is one of the most robust and consistently replicated in the literature (Swanson et al., 2004). This finding has been interpreted as an alerting attention deficit (Nigg, 2006) or as an overall self-regulatory deficit (Douglas, 1999). The present review extends the understanding of this phenomenon by demonstrating that children with ADHD were most adversely affected during incongruent conditions. This suggests that whether or not there is an overall general alerting deficit, the ADHD group had even more difficulty recruiting the attentional resources required for accurate and consistent performance when burdened with the additional demands of resolving interference.

The present results have important implications for understanding the attentional processing capabilities of children with ADHD within the Attention Network Theory. It has been predicted that executive attention (i.e., interference control) and alerting attention are impaired in ADHD (Berger & Posner, 2000; Swanson et al., 1998). The results of the present review are consistent with the prediction that executive attention is less efficient in children with ADHD, at least under certain task conditions. We are only aware of two published studies (Booth et al., 2007; Konrad et al., 2006) that directly tested these hypotheses using variants of the Attention Network Test (ANT). Booth et al. used the child version of this task and found no significant differences in executive attention between groups. In contrast, the study by Konrad et al. used the more difficult and less visually stimulating adult version of the task. In this case, children with ADHD had significantly larger congruency effects relative to the TD group. It is possible that task-related characteristics (e.g., differences in task difficulty, stimuli, feedback, and duration) may have contributed to these discrepant results. It will be important for future research to replicate these results using the adult version of the ANT with a larger sample of children with ADHD Combined Type (ADHD-C) and ADHD Predominately Inattentive Type (ADHD-I).

When collapsed across studies, clear evidence for an interference control deficit in children with ADHD was found; however, several of the individual studies reported statistically nonsignificant results. This is a common finding in the broader literature on selective attention task performance of children with ADHD (Douglas, 1999). For example, overall group differences in RT are frequently found on various tasks of selective attention, but the group interaction effect of interest often fails to reach significance. As noted by Douglas (1999) "interactions are difficult to obtain because they depend on demonstrating differences between differences. The problem is further exacerbated when the subject samples are small or when there is high variability in the data" (p. 114). Many of the studies included in the present review had large variability in the data (particularly in the ADHD group) and suffered from small sample sizes. This allows for the strong possibility that individual studies' null results were due to low power rather than a true lack of a difference between groups. These issues are particularly problematic when an expected effect is in the small to medium range (Norman & Streiner, 2000), which is likely the case for differences in interference control (Nigg, 2006).

Potential Moderating Variables

An important issue that remains outstanding in the literature is the effect of potential moderating variables on the interference control abilities of children with ADHD. Based on the research to date, it has been argued that it is not currently clear that group differences in interference control are independent of intelligence or unrelated to task parameters such as task difficulty (Nigg, 2006). Due to the modest number of studies available, it was not feasible to include only the studies that met predetermined methodological criteria on to conduct a systematic evaluation of the potential effects of one or more of these variables. Consequently, the most pressing methodological issues requiring future research are highlighted here.

Subtype of ADHD. Increasingly, research has sought to determine whether children with different *DSM-IV* subtypes of ADHD show different neurocognitive profiles (Barkley, 2006). The majority of research has focused on neuropsychological task performance in children with ADHD-C (Nigg, 2005). Much less research has compared the performance of children with ADHD-C and ADHD-I on these measures (Booth et al., 2007). The general finding appears to be that children with ADHD-I perform more poorly than TD children but they do not consistently differ from children with ADHD-C on executive function (EF) tasks (e.g., Chhabildas, Pennington, & Willcutt, 2001; Houghton et al., 1999). This has led to the hypothesis that subtype-related executive control differences may be a matter of degree rather than kind (Nigg, 2006). The evidence, however, is not entirely consistent. For example, Lockwood, Marcotte, and Stern (2001) found a differential pattern of EF deficits between children with ADHD-C and ADHD-I. In that study, children with ADHD-C performed more poorly on measures of executive motor planning, switching, and inhibition than those with ADHD-I.

Subtype information was reported in eight of the studies included in this review but was only analyzed as an independent variable in one (i.e., Booth et al., 2007). Three studies only included children with ADHD-C while five included children with varying proportions of the three subtypes of ADHD. To date, very little research has specifically examined subtype differences in interference control. Consistent with the broader EF literature, children with ADHD-C and ADHD-I tend to perform more poorly on interference control tasks than TD children tend not to differ from each other (e.g., Martel, Nikolas, & Nigg, 2007; Pasini, Paloscia, Alessandrelli, Poririo, & Curatolo, 2007; Nigg, Blaskey, Huang-Pollock, & Rappley, 2002), although others have found no differences among the three groups (e.g., Houghton et al., 1999). To the best of our knowledge, only one study (Booth et al., 2007) has examined subtype differences in interference control using the Flanker task from the child ANT. In this case, no differences were found between children with ADHD-I and ADHD-C. Replication of this result as well as extension to a Simon task and other less visually stimulating versions of the Flanker task will be important for future research.

Comorbidity. Researchers have increasingly highlighted the importance of controlling for comorbid disorders when examining cognitive task performance in children with ADHD (Oosterlaan et al., 1998; Tannock, 2002). It is well known that ADHD very frequently co-occurs with other mental health disorders, especially in clinic-referred populations. For example, Wilens et al. (2002) found that 80% of clinic-referred school-aged children with ADHD were also diagnosed with at least one other comorbid disorder. The disruptive behavior disorders of Oppositional Defiant Disorder (ODD) and Conduct Disorder (CD) are the disorders most frequently comorbid with ADHD (Angold, Costello, & Erkanli, 1999). Estimates of comorbidity between ADHD and ODD/CD range from 45% to 84% in community and clinical samples (Barkley, 2006). There is some evidence to suggest that executive deficits are related to ADHD and not to comorbid ODD/CD (Sergeant et al., 2002; Tannock, 2002), but this has not yet been examined on tasks measuring interference control.

Learning disabilities also frequently co-occur with ADHD, with estimates ranging from 20% to 25% when more stringent diagnostic criteria for LD are used (Spencer, Biederman, & Mick, 2007). Children with ADHD and LD have also been found to display a wider range of neuropsychological impairments (Nigg, Hinshaw, Carte, & Treuting, 1998; Willcutt, Pennington, Olson, Chhabildas, & Hulslander, 2005) and a greater degree of impairment on neuropsychological tasks than children with ADHD alone (Jakobson & Kikas, 2007; Seidman, Biederman, Valera, Monuteaux, Doyle, & Faraone, 2006). The majority of the research that has examined the neuropsychological functioning of children with ADHD and LD has focused specifically on the LD subtype of a Reading Disability (RD). It has been suggested that children with ADHD and RD show the impairments associated with ADHD as well as those associated with RD (e.g., Nigg et al., 1998; Rucklidge & Tannock, 2002; Willcutt et al., 2005). Nine of the studies included in this review provided at least some information about co-occurring disorders but none specifically examined the potential impact of comorbid ODD/CD or LD on the performance of Flanker or Simon tasks. Future research could examine this question by including groups of children with ADHD with and without a comorbid condition (e.g., ADHD with ODD vs. ADHD without ODD) or by including rigorously diagnosed ODD-only and/or LD-only groups that have been carefully screened to ensure the absence of comorbid ADHD.

Variability of Performance. Very recently, the topic of intrasubject variability (ISV) in ADHD has been of interest. The finding of greater variability in response time in children with ADHD has been frequently reported in studies of neurocognitive functioning in this population (e.g., C. Klein, Wendling, Huettner, Ruder & Peper, 2006; Swanson et al., 2004). It has been suggested that increased ISV is the result of a deficit in self-regulatory processes including those associated with the regulation of attention, inhibition, and executive control (Douglas, 1999). As such, some researchers have suggested that indicators of ISV may be useful for discriminating between children with and without ADHD (e.g., C. Klein et al., 2006; Leth-Steensen, Elbaz, Douglas, 2000). There are a number of different methods available to quantify ISV ranging from relatively straightforward (e.g., within-participant standard deviation of RT; *SD*) to more complex (e.g., Ex-Gaussian analysis of RT distributions). The vast majority of studies included in this review provided information about intragroup variability (i.e., *SD* of the group *M*) but did not include information about intraindividual variability (i.e., within participant, within condition *SD*s). This made it impossible to compare the two groups on indicators of ISV across studies. It would be of considerable benefit for future research on this topic to report measures of ISV. One relatively simple statistic that could be reported is the coefficient of variation. This is calculated by dividing the within participant, within condition *SD* by the within participant, within condition *M*. This measure provides an estimate of variability that controls for the fact that, when all other things are equal, the *SD* for a variable increases with the mean.

Motivational and Environmental Factors. Several theorists have suggested that ADHD is associated with an atypical sensitivity to reinforcement contingencies such as reward or punishment (e.g., Aase & Sagvolden, 2006; Douglas, 1999; Sergeant et al., 2002; Sonuga-Barke, 2002). Reward contingencies have been shown to improve the experimental task performance of both children with ADHD and TD children, but this effect may be more pronounced in children with ADHD (Luman, Oosterlaan, & Sergeant, 2005). One study found no effects of monetary reward and punishment on Flanker task performance (Crone et al., 2003). The other studies did not mention any information about this variable. Thus, it is possible that reinforcement contingencies moderated individual study results. Researchers are encouraged to report the reinforcement contingencies used (if any) during testing.

Others have hypothesized that children with ADHD experience chronic under-arousal and therefore require higher levels of environmental stimulation (e.g., brighter colors, sounds, novelty) than TD children to reach optimal levels of performance on tasks (Zentall, 1975; Zentall & Zentall, 1983). The majority of studies reviewed here used traditional Flanker or Simon experimental paradigms (e.g., black arrows on a gray background, as in the adult ANT). In contrast, Booth et al. (2007) used the child ANT that included animated yellow fish flankers, a more colorful background, and auditory feedback after each response. It is possible that the increased stimulation produced by the child ANT enhanced the performance of the ADHD group so that it was indistinguishable from the TD group. Future research could evaluate this possibility by comparing the performance of children with ADHD and TD children on the child and adult versions of the ANT.

Limitations

Some important limitations of this review must be addressed. First, the absence, so far, of a large number of studies on the topic coupled with the significant heterogeneity in terms of sample and task characteristics made it impossible to conduct a more rigorous meta-analytic review. Nevertheless, this review provides a summary of the research to date on interference control in this population and highlights the outstanding issues still to be addressed. This will allow for a more rigorous meta-analytic review to be conducted in the future. Secondly, in a small minority of the studies we were unable to obtain a few of the necessary values. This resulted in fewer data points being included in the graphical and Wilcoxon Signed Rank analyses. One recommendation that follows from this is that authors should report their findings in a manner that will permit future researchers to conduct systematic meta-analyses. For Flanker and Simon data to add to the body of literature on interference control in a meaningful way, future studies should report mean RT, percent accuracy, within participant, within condition SD of RT, and other measures of ISV (e.g., coefficient of covariation) separately by trial type and group.

Conclusion

The present review summarized the existing literature that has evaluated the interference control abilities of children with ADHD using Flanker and Simon tasks. Specific performance disadvantages were found in the ADHD group in terms of RT, errors made, and IE on incongruent relative to congruent trials, indicating weaker interference control in this group relative to children without this disorder. These results suggest that an executive control deficit in this population may encompass not only response suppression but also interference control. This review further highlights a number of important methodological and procedural issues that should be addressed in future research on this topic.

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REFERENCES

Aase, H., & Sagvolden, T. (2006). Infrequent, but not frequent, reinforcers produce more variable responding and deficient sustained attention in young children with attention-deficit/hyperactivity disorder (ADHD). *Journal of Child Psychology and Psychiatry, 47,* 457–471.

Akhtar, N., & Enns, J. T. (1989). Relations between covert orienting and filtering in the development of visual attention. *Journal of Experimental Child Psychology, 48,* 315–334.

American Psychiatric Association. (1980). *Diagnostic and statistical manual of mental disorders, Third Edition*. Washington, DC: American Psychiatric Press.

American Psychiatric Association. (1987). *Diagnostic and statistical manual of mental disorders, Third Edition-Revised*. Washington, DC: American Psychiatric Press.

American Psychiatric Association. (1994). *Diagnostic and statistical manual of mental disorders, Fourth Edition*. Washington, DC: American Psychiatric Press.

Angold, A., Costello, E. J., & Erkanli, A. (1999). Comorbidity. *Journal of Child Psychology and Psychiatry, 40,* 57–87.

Barkley, R. A. (1997). Behavioral inhibition, sustained attention, and executive functions: Constructing a unifying theory of ADHD. *Psychological Bulletin, 121,* 65–94.

Barkley, R. A. (1999). Response inhibition in attention-deficit hyperactivity disorder. *Mental Retardation and Developmental Disabilities Research Reviews, 5,* 177–184.

Barkley, R. A. (2006). *Attention-deficit hyperactivity disorder: A handbook for diagnosis and treatment, 3rd Edition*. New York: The Guilford Press.

Berger, A., & Posner, M. I. (2000). Pathologies of brain attentional networks. *Neuroscience and Biobehavioral Reviews, 24,* 3–5.

Bjorklund, D. F., & Harnishfeger, K. K. (1990). The resources construct in cognitive development: Diverse sources of evidence and a theory of inefficient inhibition. *Developmental Review, 10,* 48–71.

*Booth, J. E., Carlson, C. L., & Tucker, D. M. (2007). Performance on a neurocognitive measure of alerting differentiates ADHD combined and inattentive subtypes: A preliminary report. *Archives of Clinical Neuropsychology, 22,* 423–432.

Botvinick, M., Braver, T. S., Yeung, N., Ullsperger, M., Carter, C. S., & Cohen, J. D. (2004). Conflict monitoring: Computational and empirical studies. In M. I. Posner (Ed.), *The cognitive neuroscience of attention* (pp. 91–102). New York: Guilford Press.

Bush, G., Frazier, J. A., Rauch, S. L., Seidman, L. J., Whalen, P. J., Jenike, M. A., et al., (1999). Anterior cingulate cortex dysfunction in attention deficit/hyperactivity disorder revealed by fMRI and the counting stroop. *Biological Psychiatry, 45,* 1542–1552.

Castellanos, F. X., Sonuga-Barke, E. J. S., Scheres, A., Di Martino, A., Hyde, C., & Walters, J. (2005). Varieties of attention deficit/hyperactivity disorder-related intra-individual variability. *Biological Psychiatry, 57,* 1416–1423.

Chhabildas, N., Pennington, B. F., & Willcutt, E. G. (2001). A comparison of the cognitive deficits in the DSM-IV subtypes of ADHD. *Journal of Abnormal Child Psychology, 29,* 529–540.

Christie, J. (1995). *Familiarity and attention: Does what we know affect what we notice?* Master's Thesis, Dalhousie University, Halifax, NS, Canada.

Cornoldi, C., Marzocchi, G. M., Belotti, M., Caroli, M. G., De Meo, T., & Braga, C. (2001). Working memory interference control deficit in children referred by teachers by ADHD symptoms. *Child Neuropsychology, 7,* 230–240.

*Crone, E. A., Jennings, J. R., & van der Molen, M. W. (2003). Sensitivity to interference and response contingencies in attention deficit/hyperactivity disorder. *Journal of Child Psychology and Psychiatry, 44,* 214–226.

Douglas, V. I. (1999). Cognitive control processes in attention deficit/hyperactivity disorder. In H. C. Quay & A. E. Hogan (Eds.), *Handbook of disruptive behavior disorders* (pp. 105–138). New York: Kluwer Academic/Plenum Publishers.

*Dreschler, R., Brandeis, D., Foldenyi, M., Imhof, K., & Steinhausen, H. C. (2005). The course of neuropsychological functions in children with attention-deficit/hyperactivity disorder from late childhood to early adolescence. *Journal of Child Psychology and Psychiatry, 46,* 824–836.

Eriksen, B. A., & Eriksen, C. W. (1974). Effects of noise letters upon the identification of a target letter in a nonsearch task. *Perception and Psychophysics, 16,* 143–149.

Fan, J., Flombaum, J. I., McCandliss, B. D., Thomas, K. M., & Posner, M. I. (2003). Cognitive and brain consequences of conflict. *Neuroimage, 18,* 42–57.

Fan, J., McCandliss, B. D., Sommer, T., Raz, A., & Posner, M. (2002). Testing the efficiency and independence of attentional networks. *Journal of Cognitive Neuroscience, 14,* 340–347.

Friedman, N. P., & Miyake, A. (2004). The relations among inhibition and interference control functions: A latent variable analysis. *Journal of Experimental Psychology: General, 133,* 101–135.

Hanisch, C., Konrad, K., Gunther, T., & Herpertz-Dahlmann, B. (2004). Age dependent neuropsychological deficits and effects of methylphenidate in children with attention-deficit/hyperactivity disorder: A comparison of pre- and grade-school children. *Journal of Neural Transmission, 111,* 865–881.

Harnishfeger, K. K. (1995). The development of cognitive inhibition: Theories, definitions, and research evidence. In F. N. Dempster & C. J. Brainerd (Eds.), *Interference and inhibition in cognition* (pp. 175–204). San Diego, CA: Academic Press.

Homack, S., & Riccio, C. A. (2004). A meta-analysis of sensitivity and specificity of the Stroop Color and Word Test with children. *Archives of Clinical Neuropsychology, 19,* 725–743.

Hooks, K., Milich, R., & Lorch, E. P. (1994). Sustained and selective attention in boys with attention deficit hyperactivity disorder. *Journal of Clinical Child Psychology, 23,* 69–77.

Houghton, S., Douglas, G., West, J., Whiting, K., Wall, M., Langsford, S., et al., (1999). Differential patterns of executive function in children with attention-deficit hyperactivity disorder according to gender and subtype. *Journal of Child Neurology, 14,* 801–813.

Jakobson, A., & Kikas, E. (2007). Cognitive functioning in children with and without attention-deficit/hyperactivity disorder with and without comorbid learning disabilities. *Journal of Learning Disabilities, 40,* 194–202.

*Jonkman, L. M., Kemner, C., Verbaten, M. N., Van Engeland, H., Kenemans, J. L., Camfferman, G., Buitelaar, J. K., & Koelega, H. S. (1999). Perceptual and response interference in children with attention-deficit hyperactivity/disorder, and the effects of methylphenidate. *Psychophysiology, 36,* 419–429.

Jonkman, L. M., van Melis, J. J. M., Kemner, C., & Markus, C. R. (2007). Methylphenidate improves deficient error evaluation in children with ADHD: An event-related brain potential study. *Biological Psychology, 76,* 217–229.

Klein, C., Wendling, K., Huettner, P., Ruder, H., & Peper, M. (2006). Intra-subject variability in attention-deficit hyperactivity disorder. *Biological Psychiatry, 60,* 1088–1097.

Klein, R. M., Christie, J. J., & Ivanoff, J. (2004). Graphical and other methods for representing the speed and accuracy of performance. Presented at the 45th annual meeting of the Psychonomic Society, Minneapolis, MN.

Konrad, K., Neufang, S., Fink, G., & Herpertz-Dahlman, B. (2007). Long term effects of methylphenidate on neural networks associated with executive attention in children with ADHD: Results from a longitudinal functional MRI study. *Journal of the American Academy of Child and Adolescent Psychiatry, 46,* 1633–1641.

*Konrad, K., Neufang, S., Hanisch, C., Fink, G., & Herpertz-Dahlmann, B. (2006). Dysfunctional attentional networks in children with attention-deficit/hyperactivity disorder: Evidence from an event-related functional magnetic resonance imaging study. *Biological Psychiatry, 59,* 643–651.

Lansbergen, M. M., Kenemans, J. L., & van Engeland, H. (2007). Stroop interference and attention-deficit/hyperactivity disorder: A review and meta-analysis. *Neuropsychology, 21,* 251–262.

Leth-Steensen, C., King Elbaz, Z., & Douglas, V. I. (2000). Mean response times, variability, and skew in the responding of ADHD children: A response time distributional approach. *Acta Psychologica, 104,* 167–190.

Lockwood, K. A., Marcotte, A. C., & Stern, C. (2001). Differentiation of attention-deficit/ hyperactivity disorder subtypes: Application of a neuropsychological model of attention. *Journal of Clinical and Experimental Neuropsychology, 23,* 317–330.

Logan, G. D. (1994). On the ability to inhibit thought and action: A user's guide to the stop signal paradigm. In D. Dagenbach & T. H. Carr (Eds.), *Inhibitory processes in attention, memory, and language* (pp. 189–239). San Diego: Academic Press.

Lu, C. H. & Proctor, R. W. (1995). The influence of irrelevant location information on performance: A review of the simon and spatial stroop effects. *Psychonomic Bulletin and Review, 2,* 174–207.

Luman, M., Oosterlaan, J., & Sergeant, J. A. (2005). The impact of reinforcement contingencies on AD/HD: A review and theoretical appraisal. *Clinical Psychology Review, 25,* 183–213.

Martel, M., Nikolas, M., & Nigg, J. T. (2007). Executive function in adolescents with ADHD. *Journal of the American Academy of Child and Adolescent Psychiatry*, *46*, 1437–1444.

*McLaughlin, E. M. (2002). Cognitive Inhibition in children with attention deficit hyperactivity disorder (Doctoral dissertation, Dalhousie University, 2002). *Dissertation Abstracts International, 64,* 426.

Mullane, J. C., & Klein, R. M. (2008). Visual search by children with and without ADHD. *Journal of Attention Disorders, 12,* 44–53.

Nigg, J. T. (2000). On inhibition/disinhibition in developmental psychopathology: Views from cognitive and personality psychology and a working inhibition taxonomy. *Psychological Bulletin, 126,* 220–246.

Nigg, J. T. (2001). Is ADHD a disinhibitory disorder? *Psychological Bulletin, 127,* 571–598.

Nigg, J. T. (2005). Neuropsychologic theory and findings in attention-deficit/hyperactivity disorder: The state of the field and salient challenges for the coming decade. *Biological Psychiatry, 57,* 1424–1435.

Nigg, J. T. (2006). *What causes ADHD: Understanding what goes wrong and why*. New York: The Guilford Press.

Nigg, J. T., Blaskey, L., Huang-Pollock, C., & Rappley, M. D. (2002). Neuropsychological executive functions and ADHD DSM-IV subtypes. *Journal of the American Academy of Child and Adolescent Psychiatry, 41,* 59–66.

Nigg, J. T., Hinshaw, S. P., Carte, E. T., & Treuting, J. (1998). Neuropsychological correlates of childhood attention-deficit/hyperactivity disorder: Explainable by comorbid disruptive behaviour or reading problems? *Journal of Abnormal Psychology, 107,* 468–480.

Norman, G. R., & Streiner, D. L. (2000). *Biostatistics, second edition*. Hamilton: B.C. Decker Inc.

Oosterlaan, J., Logan, G. D., & Sergeant, J. A. (1998). Response inhibition in AD/HD, CD, comorbid AD/HD+CD, anxious, and control children: A meta-analysis of studies with the stop task. *Journal of Child Psychology and Psychiatry, 39,* 411–425.

Pasini, A., Paloscia, C., Alessandrelli, R., Porfirio, M. C., & Curatolo, P. (2007). Attention and executive functions profile in drug naïve ADHD subtypes. *Brain and Development, 29,* 400–408.

Pennington, B. F., & Ozonoff, S. (1996). Executive functions and developmental psychology. *Journal of Child Psychiatry, 37,* 51–87.

Posner, M. I., & DiGirolamo, G. J. (1998). Executive attention: Conflict, target detection, and cognitive control. In Parasuraman (Ed.), *The attentive brain*. (pp. 401–423). Cambridge, MA: MIT Press.

Posner, M. I., & Petersen, S. E. (1990). The attention systems of the human brain. *Annual Review of Neuroscience, 13,* 25–42.

Posner, M. I., & Rothbart, M. K. (2007). *Educating the human brain*. Washington, DC: American Psychological Association.

Ridderinkhof, K. R., Scheres, A., Oosterlaan, J., & Sergeant, J. A. (2005). Delta plots in the study of individual differences: New tools reveal response inhibition deficits in AD/HD that are eliminated by methylphenidate treatment. *Journal of Abnormal Psychology, 114,* 197–215.

Ridderinkhof, K. R., & van der Stelt, O. (2000). Attention and selection in the growing child: Views derived from developmental psychophysiology. *Biological Psychology, 54,* 55–106.

Rosenthal, R. (1991). *Meta-analytic procedures for social research*. Newbury Park, CA: Sage.

Rucklidge, J. J., & Tannock, R. (2002). Neuropsychological profiles of adolescents with ADHD: Effects of reading difficulties and gender. *Journal of Child Psychology and Psychiatry, 43,* 988–1003.

*Scheres, A., Oosterlaan, J., Guerts, H., Morein-Zamir, S., Meiran, N., Schut, R., et al., (2004). Executive functioning in boys with ADHD: Primarily an inhibition deficit? *Archives of Clinical Neuropsychology, 19,* 569–594.

Scheres, A., Oosterlaan, J., Swanson, J., Morein-Zamir, S., Meiran, N., Schut, H., et al., (2003). The effect of methylphenidate on three forms of response inhibition in boys with AD/HD. *Journal of Abnormal Child Psychology, 31,* 105–120.

Seidman, L. J., Biederman, J., Valera, E. M., Monuteaux, M. C., Doyle, A. E., & Faraone, S. V. (2006). Neuropsychological functioning in girls with attention-deficit/hyperactivity disorder with and without learning disabilities. *Neuropsychology, 20,* 166–177.

Sergeant, J. A., Guerts, H., & Oosterlaan, J. (2002). How specific is a deficit of executive functioning for attention-deficit/hyperactivity disorder? *Behavioral Brain Research, 130,* 3–28.

Simon, J. R. (1990). The effects of an irrelevant directional cue on human information processing. In R. W. Proctor & T. G. Reeve (Eds.), *Stimulus-response compatibility: An integrated perspective* (pp. 31–36). Amsterdam: North Holland.

Sonuga-Barke, E. J. (2002). Psychological heterogeneity in AD/HD – A dual pathway model of behaviour and cognition. *Behavioural Brain Research, 10,* 29–36.

*Sparkes, S. J. (2006). The influence of ADHD and experimental context on components of attention in children (Doctoral dissertation, Dalhousie University). *Dissertation Abstracts International, 67,* 6747.

Spencer, T. J., Biederman, J., & Mick, E. (2007). Attention-deficit/hyperactivity disorder: Diagnosis, lifespan, comorbidities, and neurobiology. *Journal of Pediatric Psychology, 32,* 631–642.

Stroop, J. R. (1935). Studies of interference in serial verbal reactions. *Journal of Experimental Psychology, 18,* 643–662.

Swanson, J. M., Casey, B. J., Nigg, J., Castellano, F. X., Volkow, N. D., & Taylor, E. (2004). Clinical and cognitive definitions of attention deficits in children with attention deficit/hyperactivity disorder. In M. I. Posner (Ed.), *The cognitive neuroscience of attention* (pp. 430–445). New York: Guilford Press.

Swanson, J., Posner, M. I., Cantwell, D., Wigal, S., Crinella, F., et al. (1998). Attention-deficit/hyperactivity disorder: Symptom domains, cognitive processes and neural networks. In Parasuraman (Ed.), *The attentive brain.* (pp. 445–460). Cambridge, MA: MIT Press.

Tannock, R. (2002). Cognitive correlates of ADHD. In P. S. Jensen & J. R. Cooper (Eds.), *Attention deficit hyperactivity disorder: State of the science/best practices* (pp. 8–27). Kingston, NJ: Civic Research Institute.

Townsend, J. T., & Ashby, F. G. (1983). *Stochastic modelling of elementary psychological processes.* New York: Cambridge University Press.

*Tsal, Y., Shalev, L., & Mevorach, C. (2005). The diversity of attention deficits in ADHD: The prevalence of four cognitive factors in ADHD versus controls. *Journal of Learning Disabilities, 28,* 142–157.

*Tucha, O., Prell, S., Mecklinger, L., Bormann-Kischkel, C., Kubber, S., Linder, M., et al. (2006). Effects of methylphenidate on multiple components of attention in children with attention-deficit/hyperactivity disorder. *Psychopharmacology, 185,* 315–326.

Tucha, O., Walitza, S., Mecklinger, L., Sontag, T. A., Kubber, S., Linder, M., et al., (2006). Attentional functioning in children with ADHD – predominately hyperactive-impulsive type and children with ADHD – combined type. *Journal of Neural Transmission, 113,* 1943–1953.

*Vaidya, C. J., Bunge, S. A., Dudukovic, N. M., Zalecki, C. A., Elliott, G. R., & Gabrieli, J. D. E. (2005). Altered neural substrates of cognitive control in childhood ADHD: Evidence from functional magnetic resonance imaging. *American Journal of Psychiatry, 162,* 1605–1613.

Van Mourik, R., Oosterlaan, J., Sergeant, J. A. (2005).The stroop revisited: A meta-analysis of interference control in AD/HD. *Journal of Child Psychology and Psychiatry, 46*, 150–165.

Wilens, T. E., Biederman, J., Brown, S., Tanguay, S., Monuteaux, M. C., Blake, T. J. et al. (2002). Psychiatric comorbidity and functioning in clinically referred preschool children and school-age youths with ADHD. *Journal of the American Academy of Child and Adolescent Psychiatry, 41,* 262–268.

Willcutt, E. G., Doyle, A. E., & Nigg, J. T., Faraone, S.V., & Pennington, B.F.(2005). Validity of the executive function theory of Attention-Deficit/Hyperactivity Disorder: A meta-analytic review. *Biological Psychiatry, 57,* 1336–1346.

Willcutt, E. G., Pennington, B. F., Olson, R. K., Chhabildas, N., & Hulslander, J. (2005). Neuropsychological analyses of comorbidity between reading disability and attention deficit hyperactivity disorder: In search of the common deficit. *Developmental Neuropsychology, 27,* 35–78.

Zang, Y., Zhen, J., Xu-Chu, W., Zhang, L., Zeng, Y., Yang, L., et al. (2005). Functional MRI in attention-deficit/hyperactivity disorder: Evidence for hypofrontality. *Brain & Development, 27,* 544–550.

Zimmermann, P., & Fimm, B. (2002). A test battery for attentional performance. In M. Leclercq & P. Zimmermann (Eds.), *Applied neuropsychology of attention: Theory, diagnosis and rehabilitation* (pp. 110–151). Hove, UK: Psychology Press.

Zentall, S. S. (1975). Optimal stimulation as theoretical basis of hyperactivity. *American Journal of Orthopsychiatry, 45,* 549–563.

Zentall, S. S., & Zentall, T. R. (1983). Optimal stimulation: A model of disordered activity and performance in normal and deviant children. *Psychological Bulletin, 94,* 446–471.

*Studies with an asterisk were included in the review.